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Real-Time Signal Optimisation Using Machine Learning and Computer Vision

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ABSTRACT

Traffic flows in urban areas, especially during peak hours, lead to significant economic and environmental issues. There are many factors that contribute to these congestions, one of which is poorly optimized traffic signals, causing delays and congestion. In this project, we aim to apply data-driven, innovative methods to address traffic problems, particularly those related to signal optimization. Image processing was performed to count vehicles, and the results were compared with manual counting to ensure accuracy. Signal design was then performed based on the extracted volumetric data using a genetic algorithm known as Python-NEAT. This neural network-based algorithm performed real-time signal optimization by feeding it data from the image processing model. Finally, the performance of this method was evaluated by comparing it with a fixed-time signal design methodology. The major findings indicate that the image processing model achieves a reasonable accuracy of 72% in vehicle detection, with potential for further improvement. Furthermore, the implementation of the neural network-based algorithm for real-time signal optimization, compared with the fixed-time method, significantly reduced queue lengths by 50%. In conclusion, this study highlights the potential for intelligent transportation systems to contribute to sustainable urban development goals by reducing congestion and improving urban mobility through reduced delays, ultimately leading to lower emissions and promoting sustainability.